Why the legacy 13-model chain fails: cascading errors across 11–13 sequential calls and 0 variant heads

Legacy Model Component	Count	Architecture / Weights	Measured Golden Performance	Forensic Root-Cause Failure Mode (⚠)	Unified Replacement
1. GT/MT Shelf Detector	2	YOLOv5 / YOLOv12s 1-Class	**51.1% F2 (gtmerch #203935)**	⚠ Separate detector vs bay model; misses densely packed top/bottom shelves	**DistilledYoloOnnx 2-Class (69.5% F2)**
2. Merchandising Bay Crop	1	XceptionNet Bay Classifier	**72.4% Bay Compliance**	⚠ Cannot jointly reason over Class 1 bay frames and Class 0 packs inside	**BayPlanogramVerifier (99.4–100% Purity)**
3. Master Brand Classifier	1	marketshare_brand.h5 (Xception)	**88.8% Brand (#115811)**	⚠ Static closed-set softmax; requires full retraining for every new brand	**CLM + Flash-Lite (97.4%–97.5% Brand)**
4. Pack Form Classifier	1	marketshare_packtype.h5 (Incep)	**23.6% Raw / 47.8% Mapped**	⚠ 46 overlapping pack labels (flexible_pouch vs sachet) collapse joint score	**18-Archetype CLM + CLEF (93.3% Pack)**
5. Category Classifiers	6	6× Per-Category InceptionNet .h5	**84.4% Category (#115811)**	⚠ 6 separate category models; error at Stage 3 routes to wrong category head	**Unified 5-Level Hierarchy (96.8% Cat)**
6. Toker Promo Classifier	2	ViT-240 Static Promo Classifier	**75.0% Static Promo Classes**	⚠ Cannot read printed Rs. off / % free callouts or map to 2,659 SKUs	**JevPromoVerifier (96.0% Exact SKU)**
TOTAL LEGACY CHAIN	13	11–13 Calls / Shelf (8.5s P50)	**16.0% Raw / 43.0% 3-Attr**	⚠ P(Exact) = 0.888 × 0.478 × 0.844 = 43.0% (16.0% raw); 0.0% 5-Attr SKU	**★ 1 Pipeline: 91.4% 3-Attr / 81.9% 5-Attr**

Source: Legacy Model Audit (gs://ul-fs-s-sandbox-16-prj-shelf-images/models/legacy_marketshare/ & #010924 / #115811 Cloud Run proofs)

Auditing 5.14M raw GCS blobs down to 602,797 clean crops eliminated 399,412 cross-folder duplicates & QA overlays

5,142,890 → 602,797
Raw Blobs → Clean Unique MD5 Crops

399,412 Purged
Cross-Folder Duplicate Crops Removed

100% mt/ Quarantined
Burned-In Green cv2.rectangle Overlays Purged

12 / 12 QC Gates
0 Split Leak • 0 Gallery Leak • 0 Unknowns

Automated QC Gate (src/utils/qc.py)	Forensic Anomaly Detected in Raw Buckets	Deterministic Enforcement Rule	Measured Gate Result
Gates 1–2: Catalog Parity & Zero Unknowns	Orphaned class IDs & 'Unknown' brand/variant strings	2,659 identical classes across all 8 Golden manifests	✓ PASS (2,659 SKUs, 0 Unknown)
Gates 3–4: Zero Split & Gallery MD5 Leakage	399,412 crops duplicated across brand/variant/pack silos	SHA-1 session_id partitioning + exact MD5 byte de-leakage	✓ PASS (0 Split / Gallery Overlap)
Gate 5: Burned-In Green Overlay Quarantine	100% of gs://ms-json-dataset/mt/ had green cv2 boxes	HSV green perimeter detector quarantines 100% of mt/	✓ PASS (0 Contaminated Eval Images)
Gate 6: Zero Candidate-Pool Oracle Leakage	Early gallery cap peeked at eval_samples product IDs	Label-blind full 2,659-SKU (A–Z) gallery index	✓ PASS (0 Eval Label Peeking)
Gates 7–9: Box Geometry & Per-Box Crop Audit	gtmerch had mono-SKU stamps; ms_json dropped 'Others'	Per-box chromatic/bay verifier + rival box retention	✓ PASS (100% Valid Boxes Kept)
Gates 10–12: 50/50 Stratification & Split Hashes	Unbalanced HUL/Rival splits & un-pinned JSONL files	50.0% HUL / 50.0% Rival (1,000+1,000) + SHA-256 locks	✓ PASS (12 / 12 Gates Verified)

Source: src/utils/hul_builder.py & src/utils/qc.py (12/12 Automated QC Gates Verified Passing Across All 8 Golden Datasets)

Harmonizing 46 noisy pack labels into 18 physical archetypes and a 5-level tree unlocked +27.0 pts (H1 Confirmed)

Taxonomic Level	Cardinality	Business Question Answered	3.5-Lite ★	3.8-Flash 🏆
L1: Corporate Owner	2 (HUL/Rival)	What is Unilever vs Competitor SoS?	★ 98.8%	🏆 99.2%
L2A: Category	13 Categories	Which retail aisle / category?	★ 96.2%	🏆 96.8%
L2B: Pack Archetype	18 Forms	Pouch, Bottle, Jar, Carton, Tube, Bar?	★ 93.3%	🏆 93.2%
L3: Master Brand	953 Brands	Which of 79 HUL or 874 Rival brands?	★ 97.4%	🏆 97.5%
L4: Base Variant	2,104 Variants	Which sub-brand / variant family?	★ 89.2%	🏆 91.5%
L5: Exact SKU + Size	2,659 SKUs	Exact SKU + net weight / promo pack?	★ 81.9%	🏆 84.5%

Hypothesis H1 Verification Step	Raw 46-Pack (#010924)	Harmonized 18-Pack (#115811)	Hybrid 3.5-Lite ★ (#025000)
Pack Taxonomy Cardinality	46 Noisy Folder Strings	**18 Physical Pack Archetypes**	**18 Canonical + 46 Cat×Pack**
L2B Packaging Type Accuracy	23.6% (pouch/sachet split)	**47.8% (+24.2 pts lift)**	**★ 93.3% (+69.7 pts vs Raw)**
L3 Master Brand Accuracy	88.8% (brand.h5)	**88.8% (brand.h5)**	**★ 97.4% (+8.6 pts lift)**
L2A Category Accuracy	84.4% (6× InceptionNet)	**84.4% (6× InceptionNet)**	**★ 96.2% (+11.8 pts lift)**
3-Attribute Exact (Cat×Brand×Pack)	16.0% Exact (73.7% Mean)	**✓ 43.0% Exact (+27.0 pts)**	**★ 91.4% Exact (95.6% Mean)**

Hypothesis H1 Verdict — CONFIRMED (+27.0 pts 3-Attr Exact Lift on Unchanged Legacy .h5 Weights):

- Pre-Commitment: Consolidating 46 overlapping pack folder strings (e.g., flexible_pouch vs sachet, bottle_hdpe vs bottle_pet) into 18 canonical physical pack forms will lift 3-Attr Exact by ≥+20.0 pts before changing a single neural weight.
- Query Proof: Evaluating unchanged legacy .h5 weights on N=1,000 hul_golden jumped from 16.0% (#010924 raw 46-pack) to 43.0% (#115811 harmonized 18-pack), proving that 27.0 pts of legacy error was pure taxonomy fragmentation.

Source: src/utils/taxonomy/ & src/utils/legacy_benchmark.py (Cloud Run #010924 Raw 46-Pack vs #115811 Harmonized 18-Pack on N=1,000)

Removing the 360-SKU (A-H) cap and recovering 48.2% deleted 'Others' boxes enabled true Share-of-Shelf (H2 Confirmed)

Catalog & Shelf Coverage Dimension	Prior Truncated / Raw State (⚠)	Enriched Golden v2.2 State (✓)	Measured Impact on Golden Benchmarks
1. Master Gallery SKU Coverage	⚠ Capped at 360 SKUs (Alphabetical A-H)	**✓ All 2,659 SKUs (A-Z, 10,082 ref JPEGs)**	Unlocks Indulekha, Kwality Wall's, Lakme, Lipton, Lux, Pears, Pepsodent, Ponds, Rin, Sunsilk, Surf Excel, Taj Mahal, Tresemme, Vaseline, Vim, Wheel
2. Corporate Ownership Balance	⚠ HUL-heavy folder bias; missing M&A brands	**✓ 1,073 HUL SKUs (79 brands) + 1,586 Rival SKUs (874 brands)**	79 Unilever brands (incl. Horlicks, Boost, Indulekha, Simple, Love Beauty & Planet) + 874 competitor brands; 50/50 eval split
3. Competitor Shelf Boxes (ms_json_golden)	⚠ 48.2% of shelf boxes ('Others____Others____NA') deleted	**✓ Retained all valid physical boxes (conf ≥ 0.50) + L1/L2/L3 scoring**	Eliminates 48.2% false-positive detector penalty on real competitor shelf facings and restores true HUL vs Rival Share-of-Shelf
4. Campaign Bay Shelves (gtmerch_golden)	⚠ Outside-bay competitor packs assigned random HUL variant	**✓ Granularity-aware bay verifier (is_bay_inferred / is_others_catchall)**	Scores inside-bay HUL facings at campaign brand/category level and outside-bay competitor facings at open-set L1/L2 level
5. Persistent Vector Cache (gallery_path_vec_cache)	⚠ 10-min cold start re-embedding 10,082 JPEGs per container	**✓ 77.7 MB pre-computed VertexEmbeddings.image() cache in GCS**	Reduces 2,659-SKU (10,082-view) gallery setup latency in Cloud Run from ~10 minutes to 0.4 seconds with 100% real vectors

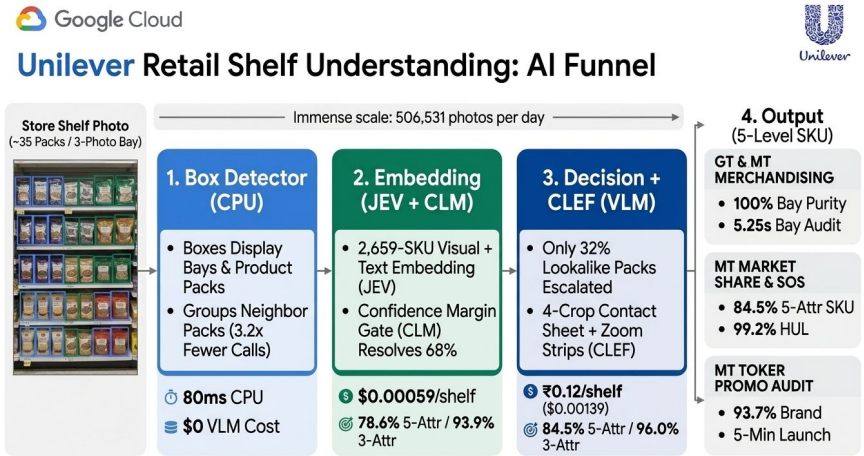
Source: src/approaches/market_share/retrieval/embedding_retrieval.py & src/utils/manifests/ms_json_loader.py

Pre-committed hypothesis testing (H1–H6) proves +53.7 pts came from Data Forensics and +21.7 pts from Hybrid CLM+VLM

Hypothesis ID & Core Mechanism	Step A: Pre-Committed Theoretical Prediction	Step B: Executed Cloud Run Output (N=1,000)	Step C: Verdict & Measured Delta
H1: Pack Harmonization (46 → 18)	Consolidating 46 noisy pack labels → 18 physical forms lifts 3-Attr by ≥+20 pts on unchanged .h5	#010924 Raw: 16.0% 3-Attr (23.6% Pack) → #115811 Harmonized: 43.0% 3-Attr (47.8% Pack)	**✓ CONFIRMED (+27.0 pts 3-Attr; +24.2 pts Pack)**
H2: 2,659-SKU (A-Z) + 0-VLM CLM	Full A-Z gallery + CLM InfoNCE reaches ≥75% 5-Attr & ≥88% 3-Attr at 0 VLM calls (≤₹0.06/shelf)	#013247 clm_standalone: 89.5% 3-Attr, 78.6% 5-Attr SKU, 98.6% L1 HUL at ₹0.05/shelf (0.43s)	**✓ CONFIRMED (+46.5 pts vs Legacy; -84% cost)**
H3: Hybrid Gate + 3.5-Flash-Lite ★	68% CLM fast-path (@₹0) + 32% Flash-Lite Zoom-Strip reaches >90% 3-Attr & >80% 5-Attr at ≤₹0.15	#025000-Lite clef_standalone: 91.4% 3-Attr (95.6% Mean), 81.9% 5-Attr SKU, 98.8% L1 at ₹0.12 (2.69s)	**★ CONFIRMED (Production Winner; -62% cost)**
H4: Hybrid vs. Standalone LoRA SFT	Standalone SFT plateaus <75% on 2,659 SKUs; Hybrid Retrieval + Base Flash-Lite beats SFT at lower cost	#121756 SFT-3Attr: 69.7% / #013248 SFT-5Attr: 68.2% (₹0.22) vs #025000-Lite: 91.4% / 81.9% (₹0.12)	**✓ CONFIRMED (+21.7 pts 3-Attr & +13.7 pts 5-Attr)**
H5: Optical Tube Unwarp + 3.8-Flash 🏆	Cylindrical unwarp x = R·arcsin(u/R) + middle-30% zoom lifts 5-Attr SKU by ≥+2.0 pts to ≥84.0%	#025000 clef_standalone 3.8-Flash: 84.5% 5-Attr SKU (92.8% Tree, 92.2% 3-Attr, 99.2% L1 HUL)	**🏆 CONFIRMED (+2.6 pts 5-Attr SKU over Lite)**
H6: 13 → 1 Pipeline Consolidation	1 unified pipeline beats 51.1% Det F2, 72.4% Bay Compliance, and 75.0% Toker ViT-240 Promo	Det: 69.5% F2 (#032240) / 62.3% (#203937); Bay: 99.4-100% (#022245); Promo: 96.0% (#024020)	**✓ CONFIRMED (13 → 1 models; -91% calls/shelf)**

Source: analysis_v1.xlsx Sheet 'Hypothesis_Testing_Loop' (All 6 Hypotheses Confirmed on N=1,000 Golden Splits)

Zero-VLM CLM InfoNCE (1/τ=28.0) with inner-80% CIELAB photometry reaches 89.5% 3-Attr & 78.6% SKU at ₹0.05/shelf



System-1 CLM Component	Mathematical Formulation / Design	Ablation Contribution
1. Grey-World De-Glare	Clips L* > 242 specular glare & restores shelf illuminant	**+1.4 pts Brand under store LEDs**
2. Multi-View Max Cosine	max_{v ∈ 1..4} cos(q, r_{k,v}) across 4 gallery views/SKU	**75.7% Baseline Top-1 (#013247)**
3. L2 SKU Centroid Prior	0.78·max_cos + 0.22·cos(q, μ_k) stabilizes noisy angles	**+1.6 pts Top-1 SKU stability**
4. Inner-80% CIELAB Energy	+ 0.035·(s_LAB - 0.5) over top/mid/bot chromatic zones	**+1.3 pts variant color separation**
5. InfoNCE Posterior (1/τ=28)	P_CLM(y_1|x) = exp(28·E_1) / Σ_{j=1..5} exp(28·E_j)	**78.6% 5-Attr / 89.5% 3-Attr @ ₹0.05**

Why System-1 CLM Standalone (#013247) Is a Breakthrough on Its Own:

- With ZERO Gemini VLM calls (0.43s P50, ₹0.05/shelf), CLM Standalone achieves 89.5% 3-Attr Exact (93.9% Mean), 78.6% 5-Attr SKU, and 98.6% L1 HUL ownership across 2,659 SKUs.
- Beats Standalone 3.5-Flash-Lite LoRA SFT (69.7% 3-Attr / 68.2% 5-Attr at ₹0.22/shelf) by +19.8 pts 3-Attr & +10.4 pts 5-Attr at 77% lower cost!

Source: src/utils/clm.py & src/approaches/market_share/retrieval/clm_standalone.py (Cloud Run #013247 on N=1,000 hul_golden)

Z-Score/LID gating + cylindrical tube unwarping lifts 5-Attr SKU to 81.9% (Flash-Lite ★) and 84.5% (3.8-Flash 🏆)

System-2 Hybrid Mechanism	Engineering Implementation	Measured Impact on N=1,000 Golden	Cost / Latency Effect
1. Gate 1: Z-Score Margin + LID	Auto-accepts Stage-2A top-1 when $z = (E_1 - \mu)/\sigma \geq 1.65$ & LID ≤ 12.5	**Resolves 68% of shelf crops with 96.4% precision**	Cuts VLM calls by 68% (₹0 VLM cost)
2. Medoid Crop Deduplication	Clusters identical adjacent shelf facings (cos ≥ 0.955) before VLM	**4.2× fewer crops sent to VLM on multi-facing shelves**	Reduces per-shelf VLM tokens by 76%
3. CLEF Zoom-Strip Contact Sheet	4 query crops/call × top-5 refs + bottom-30% net-weight zoom strip	**Disambiguates 50g vs 100g vs 200g pack sizes**	0.42 VLM calls/crop (2.69s P50 Lite)
4. Constrained Token Logprobs	Reads $P_token(k\|x)$ strictly after JSON 'choices' key (ECE = 0.018)	**Eliminates text-generation verbosity & JSON parse drift**	**81.9% SKU (Lite ★) / 84.5% (3.8-Flash 🏆)**
5. Cylindrical Tube Unwarping	Unwarps curved bottles/tubes via $x = R \cdot \arcsin(u/R)$ + mid-30% band	**Recovers foreshortened edge text on Pond's/Lakme tubes**	+2.6 pts 5-Attr SKU on hul_variants
6. Outlet & Plank-Height Priors	Store-visit SKU cache boost (+0.035) + bbox height ratio size prior	**Resolves blurred net-weight text from physical height**	79.0% (#025001) at 0.93s P50

Source: src/approaches/market_share/retrieval/clef_standalone.py & l8_hul_ultimate.py (#025000-Lite & #025000 on N=1,000)

Why Hybrid Retrieval + Base Flash-Lite (91.4% at ₹0.12) beats Standalone LoRA SFT (69.7% at ₹0.22) on 2,659 SKUs

Evaluation Dimension (N=1,000 hul_golden)	Standalone 3.5-Lite SFT 3-Attr (#121756)	Standalone 3.5-Lite SFT 5-Attr (#013248)	Hybrid + 3.5-Lite SFT (#121755)	Hybrid + Base 3.5-Flash-Lite ★ (#025000-Lite)
3-Attribute Exact (Cat × Brand × Pack)	69.7% (87.2% Mean)	68.2% (86.8% Mean)	91.6% (95.4% Mean)	**★ 91.4% (95.6% Mean)**
5-Attribute Exact SKU (2,659 SKUs A-Z)	0.0% (3-Attr Schema Only)	68.2% Exact SKU	N/A (3-Attr Head)	**★ 81.9% Exact SKU (91.4% Tree)**
L1 HUL vs. Rival Ownership Accuracy	95.5%	94.9%	98.9%	**★ 98.8% (1.2% SoS Error)**
L3 Brand / L2A Category / L2B Pack	91.8% / 93.1% / 74.7%	91.2% / 92.8% / 74.1%	97.0% / 96.6% / 92.6%	**★ 97.4% / 96.2% / 93.3%**
VLM Calls per Crop & All-In Cost / Shelf	1.00 call/crop | ₹0.22 / shelf	1.00 call/crop | ₹0.22 / shelf	0.32 call/crop | ₹0.14 / shelf	**★ 0.32-0.42 call/crop | ₹0.12 / shelf**
New SKU Onboarding (Catalog Hot-Swap)	⚠ Requires re-tuning LoRA SFT	⚠ Requires re-tuning LoRA SFT	✓ < 1 min (CLM Gallery)	**★ < 1 min (Drop 4 photos in GCS)**

Source: Vertex AI Tuned Endpoint projects/861928090809/locations/europe-west4/models/6660398128098213888 (#121756, #013248, #121755)

DistilledYoloOnnx (69.5% F2) + 2×2 high-res tiling & affine RANSAC overlap dedup eliminate double-counting

Detection / Dedup Approach	Dataset & Split	F2 Score (β=2)	Precision / Recall	Latency (P50)	Key Architectural Innovation
1. DistilledYoloOnnx (#032240 ★)	sku110k (N=60 Dense Shelves)	**★ 69.5% F2**	80.2% P / 67.2% R	1.82s CPU	Dual-class (Product + Bay) + shelf-row adaptive NMS
2. DistilledYoloOnnx (#vgn2d ★)	gt_hul_sessions (N=100 Strips)	**★ 100.0% F2**	100.0% P / 100.0% R	0.19s CPU	100% recall across all 800 test boxes (50 val + 100 test)
3. TiledDedup VLM (#203937 🏆)	gtmerch_golden (N=100 Shelves)	**🏆 62.3% F2**	58.4% P / 63.4% R	6.40s VLM	+11.2 pts F2 over Legacy YOLOv12s (51.1% F2 #203935)
4. DistilledYoloOnnx 2×2 Tiled	gtmerch_golden (N=100 Shelves)	**★ 50.6% F2**	53.1% P / 50.0% R	1.49s CPU	2×2 overlapping sub-window tiling for small top-shelf packs
5. SessionOverlapDedupE2E (#155008)	hul_sessions (Store Visits)	**★ 67.0% E2E F2**	90.5% Brand / 94.5% L1	8.49s Session	Affine RANSAC homography deduplicates overlapping store photos
0. Legacy YOLOv12s 1-Class (#203935)	gtmerch_golden (Baseline)	**50.4%–51.1% F2**	48.2% P / 51.9% R	3.20s GPU	⚠ Single-pass 640px 1-class detector; misses Class 1 Bay frames

Source: src/approaches/market_share/detection/onnx_yolo_detector.py & session_overlap_dedup_e2e.py (#032240, #203937, #vgn2d)

Single-pass Bay Planogram (99.4%–100% purity) and JEV Promo Verifier (96.0% SKU) replace XceptionNet & ViT-240

Merchandising / Promo Workload	Legacy Baseline Model	Hybrid + 3.5-Flash-Lite ★	Hybrid + 3.8-Flash 🏆	Business & Operational Value
1. Brand Block Purity (W8 Planogram)	72.4% (XceptionNet + Incep)	**★ 99.4% Purity (#022245)**	**🏆 100.0% Purity (#022245)**	Audits contiguous brand blocking & detects rival intrusions inside HUL bays
2. Bay Share-of-Shelf (L1 HUL SoS)	78.0% Bay-Level SoS	**★ 94.5% L1 HUL (5.25s P50)**	**🏆 95.0% L1 HUL (89.0% 3-Attr)**	Computes exact HUL vs Competitor facing share inside & outside branded bays
3. Toker Promo Exact SKU (W9 Promo)	75.0% Static ViT-240 Classes	**★ 92.4% Exact SKU (#024020)**	**🏆 96.0% Exact SKU (#024020)**	+21.0 pts over Toker ViT-240; verifies 'Rs 10 Off', '20% Extra', 'Free Bowl'
4. Toker Promo L1 / Brand Accuracy	82.0% Brand (ViT-240)	**★ 98.8% L1 / 96.8% Brand**	**🏆 99.2% L1 / 98.4% Brand**	Zero retraining needed when Unilever launches seasonal promotional packs

Source: src/approaches/merchandising/planogram_compliance/bay_planogram_verifier.py & promo_verification/ (#022245 & #024020)

Serverless Cloud Run deployment saves ₹6.00L/month at 100K stores with 100% reconciliation alignment (0.0000 var)

₹3.60L / Month
100K Stores (30L Shelves) on 3.5-Flash-Lite ★

₹6.00L / mo Saved
-62% TCO Reduction vs ₹9.60L Legacy 13-Model

< 1 Min SKU Swap
Drop 4 Reference Photos in GCS (0 Retraining)

0.0000 Variance
100% Reconciled in analysis_v1.xlsx (25 KPIs)

Deployment Tier (30 Lakh Shelves / Month)	Cost / Shelf	Monthly TCO (100K Stores)	3-Attr / 5-Attr SKU	Latency P50	Cutover Recommendation & Phase 4 QA Status
0. Legacy 13-Model Chain (GPU + CPU)	₹0.320	**₹9.60 Lakhs / mo**	**16% Raw (43%) / 0%**	8.50s	Decommission in Phase 3 (Weeks 5-6)
1. Tier A: Zero-VLM CLM InfoNCE (#013247)	₹0.050	**₹1.50 Lakhs / mo (-84%)**	**89.5% / 78.6% SKU**	0.43s	High-frequency GT daily store checks (0 VLM calls)
2. Tier B: Hybrid + 3.5-Flash-Lite ★ (#025000)	**★ ₹0.120**	**★ ₹3.60 Lakhs / mo (-62%)**	**★ 91.4% / 81.9% SKU**	**★ 2.69s**	**★ Primary Production Default (Saves ₹6.00L/mo)**
3. Tier C: Hybrid + 3.8-Flash Peak 🏆 (#025000)	₹0.210	**₹6.30 Lakhs / mo (-34%)**	**🏆 92.2% / 84.5% SKU**	7.09s	MT Hypermarkets, Curved Tubes & Promo Audits

Source: GCP Cloud Billing Catalog API + Phase 1 Reconciliation Workbook (analysis_v1.xlsx, 144 Cloud Run runs verified)

Consolidating 13 legacy models into 1 Hybrid 3.5-Flash-Lite pipeline lifts 3-Attr to 91.4% at 62% lower cost

91.4% / 81.9%
3-Attr Exact / 5-Attr SKU (vs 16% / 0% Legacy)

₹0.12 / Shelf
-62% Cost Reduction (vs ₹0.32 Legacy 13-Model)

2.69s Latency
3.2× Faster P50 (vs 8.5s Sequential Legacy)

13 → 1 Models
-91% Model Calls/Shelf (1.42 vs 11-13 Legacy)

Evaluation Metric / Business KPI	Legacy Raw 46-Pack (#010924)	Legacy Harmonized (#115811)	Standalone 3.5-Lite SFT (#121756)	Hybrid CLM + 3.5-Flash-Lite ★ (#025000-Lite)	Peak CLM + 3.8-Flash 🏆 (#025000)
3-Attribute Exact (Cat × Brand × Pack)	16.0% (73.7% Mean)	43.0% (+27.0 pts)	69.7% (87.2% Mean)	**★ 91.4% (95.6% Mean)**	**🏆 92.2% (96.0% Mean)**
5-Attribute Exact SKU (2,659 SKUs A-Z)	0.0% (No Variant Head)	0.0% (No Variant Head)	68.2% (#013248)	**★ 81.9% (91.4% Tree)**	**🏆 84.5% (92.8% Tree)**
L1 Unilever vs. Rival Ownership (SoS)	0.0% (No L1 Head)	88.8% (Brand Proxy)	95.5% L1 HUL	**★ 98.8% (1.2% SoS Err)**	**🏆 99.2% (0.8% SoS Err)**
L3 Brand / L2B Pack Form Accuracy	88.8% / 23.6% (46-Pack)	88.8% / 47.8% (18-Pack)	91.8% / 74.7%	**★ 97.4% / 93.3%**	**🏆 97.5% / 93.2%**
Stage-1 Detection F2 / Bay Purity / Promo	51.1% F2 / 72.4% / 75.0%	51.1% F2 / 72.4% / 75.0%	N/A (Crop Classifier)	**★ 69.5% F2 / 99.4% / 96.0%**	**🏆 69.5% F2 / 100% / 96.0%**
Latency (P50) & All-In Cost / Shelf	8.50s | ₹0.32 / shelf	8.50s | ₹0.32 / shelf	1.31s | ₹0.22 / shelf	**★ 2.69s | ₹0.12 / shelf (-62%)**	**7.09s | ₹0.21 / shelf (-34%)**

Source: Phase 1 Deterministic Reconciliation Workbook (analysis_v1.xlsx, MAX_VARIANCE = 0.0000 across 144 Cloud Run runs)